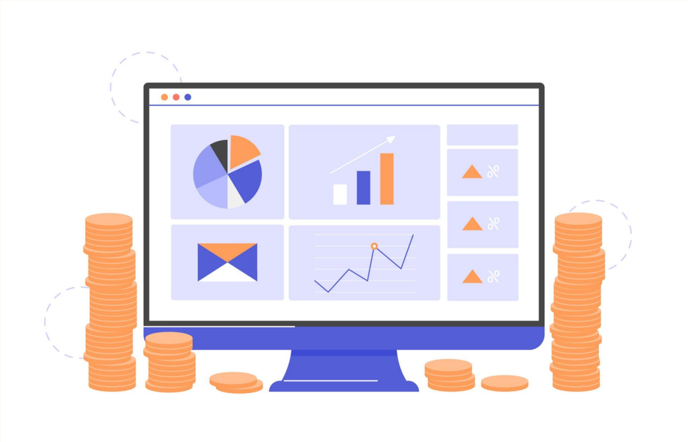
RevoFin Loan analysis



Executive Summary

Project Description :
analyzes RevoFin’s loan portfolio to evaluate overall portfolio financial health indicators, risk assessment, and customer dynamics.

Objectives :
Assess the current condition of RevoFin’s loan portfolio, Quantify outstanding loan exposure and portfolio risk, Identify high-risk and low-risk loan,Analyze customer characteristics associated with loan performance, and Provide data-driven recommendations to enhance loan underwriting and risk management.

Tools & Metrics :
Tools **:** BigQuery SQL as analysis and processing data. Tableau and spreadsheet as Visualisation.
Metrics : OS, ENR, and TKB30, Income Rate, Home status, Interest Rate, Loan Purpose.

Analysis Result :

- Credit quality improved materially after 2015.
- Home owning has better loan risk.
- Low-interest loans correlate strongly with better borrower quality and repayment capacity.
- Debt consolidation is the main growth and risk driver.

Background Of Project

As RevoFin’s digital lending portfolio grows, effective risk management becomes critical to maintain portfolio quality and revenue. This project analyzes outstanding loans and risk metrics to identify performance patterns and support better underwriting and lending decisions.

***Disclaimer**
Please note that RevoFin is a fictional company used for the purposes of this presentation. Any references to its services or data analysis methods are purely illustrative. This scenario is designed
to showcase the skills of a data analyst and is not linked to any real financial institution. Any resemblance to actual companies is coincidental.

Business Problem

As RevoFin scales its lending operations, the company faces increasing risk exposure from delinquent and defaulted loans. Limited visibility into portfolio quality and customer risk profiles makes it challenging to identify high-risk segments and optimize underwriting decisions.

Objective of analysis

- Assess the overall health and risk of the loan portfolio
- Identify high and low-risk loan cohorts using key risk metrics
- Analyze customer characteristics driving loan performance
- Provide data-driven recommendations to improve underwriting and risk management

Methodology

1

Problem Understanding

identify high-risk segments and optimize underwriting decisions.

2

Data Preparation & Data Cleaning

Filter finished loan and adjust typo.

3

Exploratory Data Analysis

Conduct Initial Investigation on dataset to define metrics and test for hypothesis

4

Data Analysis

Conduct the analysis with the metrics that decided on EDA with SQL and excel.

5

Data Visualization

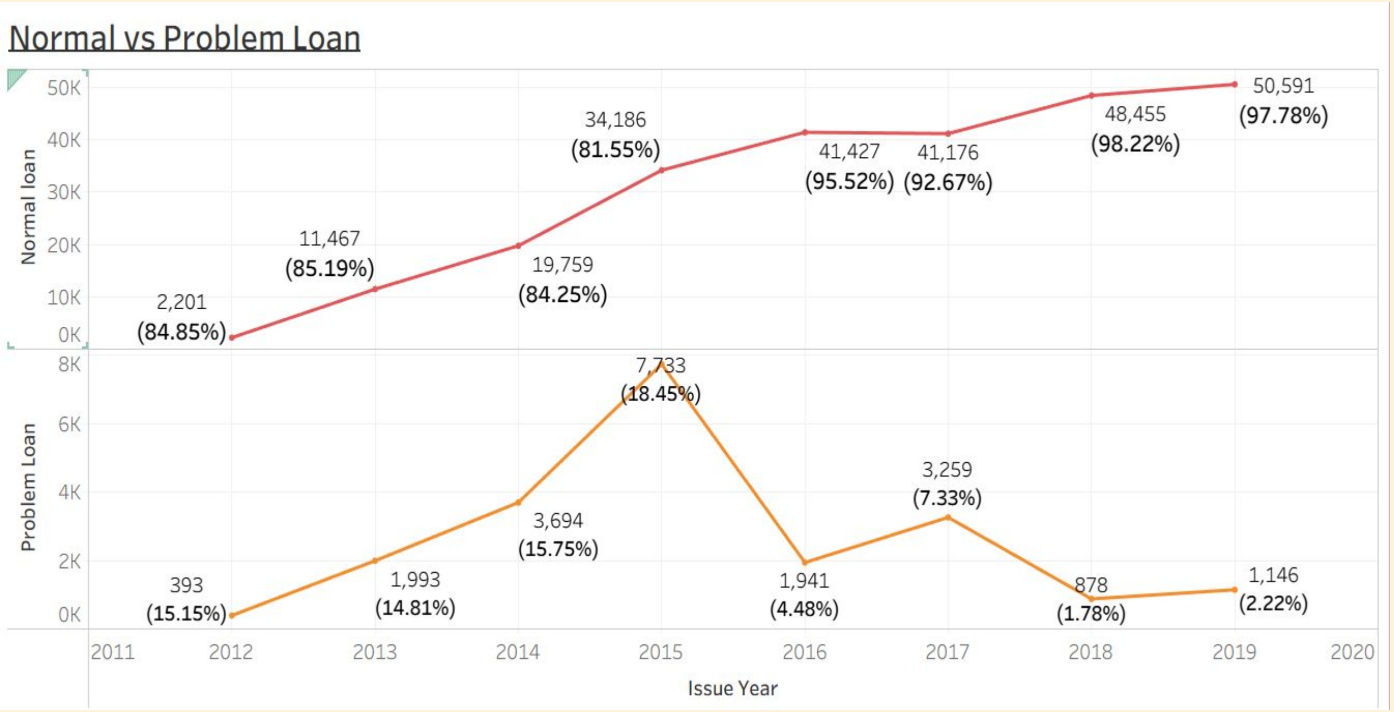
Visualize the analysis into chart, table, and dashboard by using Tableau

6

Insight & Recommendation

Use the result of data analysis to determined the insight and business recommendation to solve the problem

Risk Analysis



Normal Loan

*Loans with Current or Fully Paid status and **no delinquency beyond 30 days**.*

Problem Loan

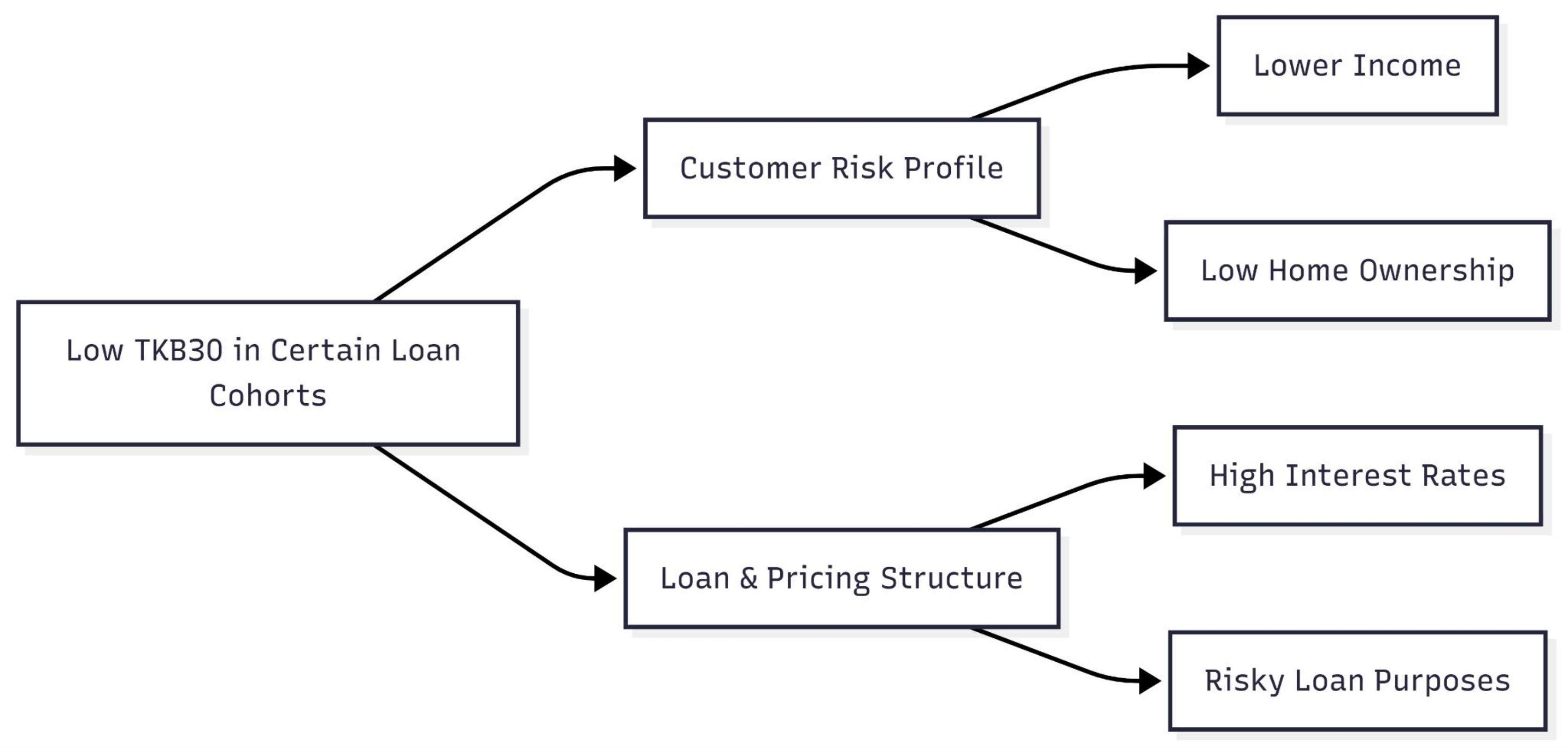
*Loans with Late (31–120 days), charged off, or Default status, represented by **TKB30**.*

Normal loans consistently represent ~82%–98% of total loans across years.

Problem loans peaked around 2014–2015, both in absolute count and percentage (up to ~18%).

After 2016, **problem loan ratios decline sharply**, reaching ~2% by 2018–2019.

Root Clause Analysis



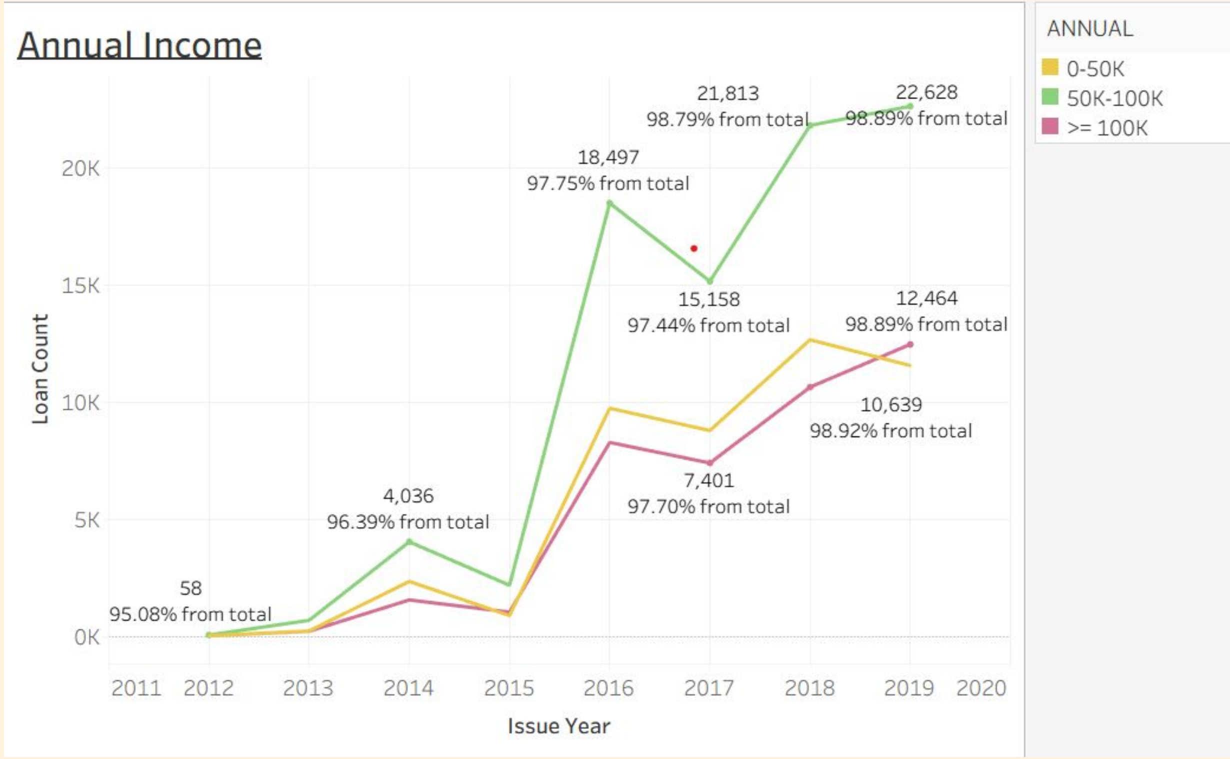
Yearly Recap

Cohort ..	
2012	OS : 5,050,568 TKB30 : 180,232 Percentage TKB30 : 7.68%
2013	OS : 52,790,273 TKB30 : 1,581,668 Percentage TKB30 : 6.59%
2014	OS : 264,864,500 TKB30 : 8,971,283 Percentage TKB30 : 6.96%
2015	OS : 196,953,468 TKB30 : 4,980,943 Percentage TKB30 : 5.87%
2016	OS : 1,082,163,361 TKB30 : 25,203,453 Percentage TKB30 : 4.59%
2017	OS : 974,851,342 TKB30 : 26,184,108 Percentage TKB30 : 5.38%
2018	OS : 1,493,358,888 TKB30 : 19,203,336 Percentage TKB30 : 2.60%
2019	OS : 1,605,521,831 TKB30 : 17,766,739 Percentage TKB30 : 2.26%

Period	Growth	TKB30 Risk	Interpretation
2012–2014	Low	High	Weak underwriting / immature risk model
2015–2017	Medium	Medium	Transition phase
2018–2019	High	Low	Optimized credit strategy

Portfolio growth accelerated after 2016 while early delinquency consistently declined, indicating materially improved underwriting quality; recent vintages combine scale with strong risk performance and should be used as the benchmark strategy.

Annual Income : Normal Active Loan



Post-2016 cohorts across all income levels show stable repayment behavior, indicating credit policy and risk screening improved after 2015.

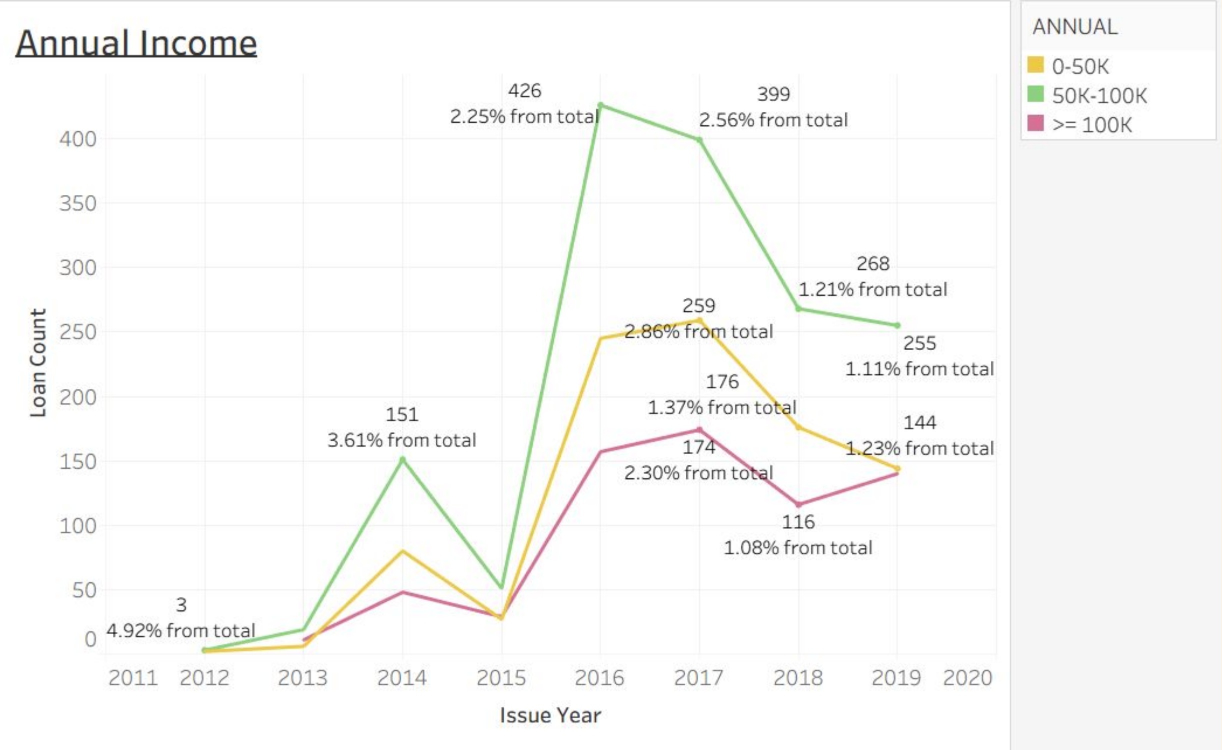
Insights :

- ➢ **50K–100K income segment consistently dominates normal loans (~97–99% performing)**, making it the backbone of healthy portfolio performance.
- ➢ Higher-income borrowers show strong and stable repayment behavior with growth potential.

Recommendation :

- ➢ Prioritize acquisition and retention in mid-income segment as core performing base.
- ➢ Expand selectively to higher-income borrowers to improve portfolio quality and stability.

Annual Income : Problem Active Loan



Pre-2016 low-to-mid income cohorts show relatively higher problem loan concentration, suggesting earlier underwriting was less selective.

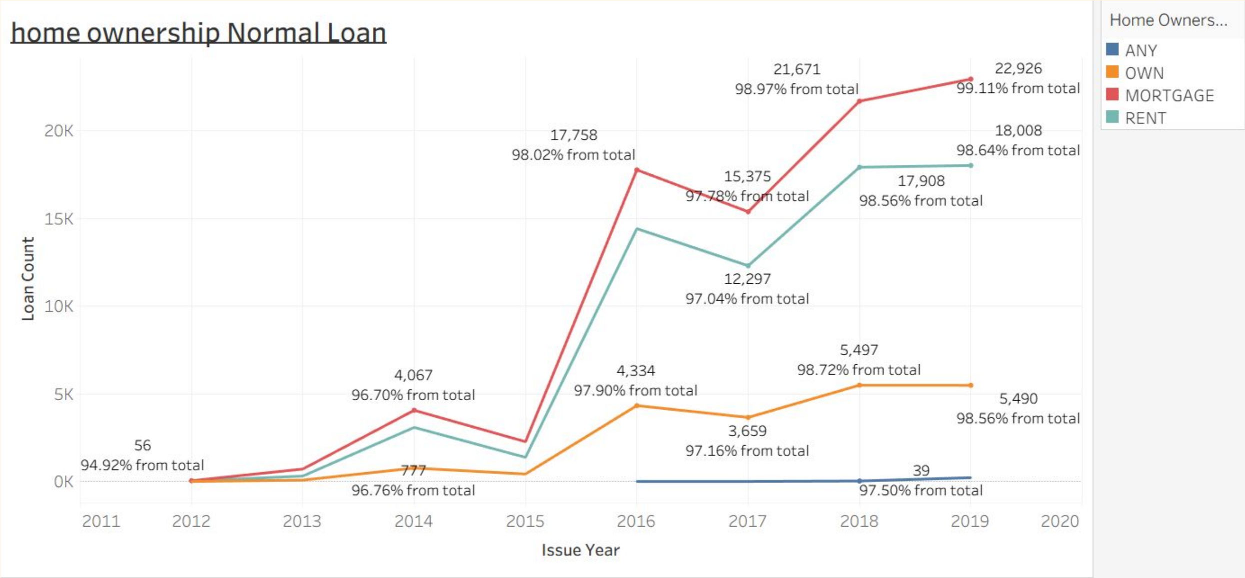
Insights :

- ➢ Problem loans are more concentrated in lower and mid-income segments, with unstable volume trends across years.
- ➢ This suggests higher sensitivity to economic pressure and repayment capacity risk in these segments.

Recommendation :

- ➢ Tighten underwriting and monitoring for lower-income borrowers.
- ➢ Introduce early warning triggers and proactive collection strategy for mid-income risk pockets.

Home Ownership - Normal Active Loan



Mortgage and Rent cohorts after 2016 consistently generate the majority of performing loans, showing strong portfolio scalability in these segments.

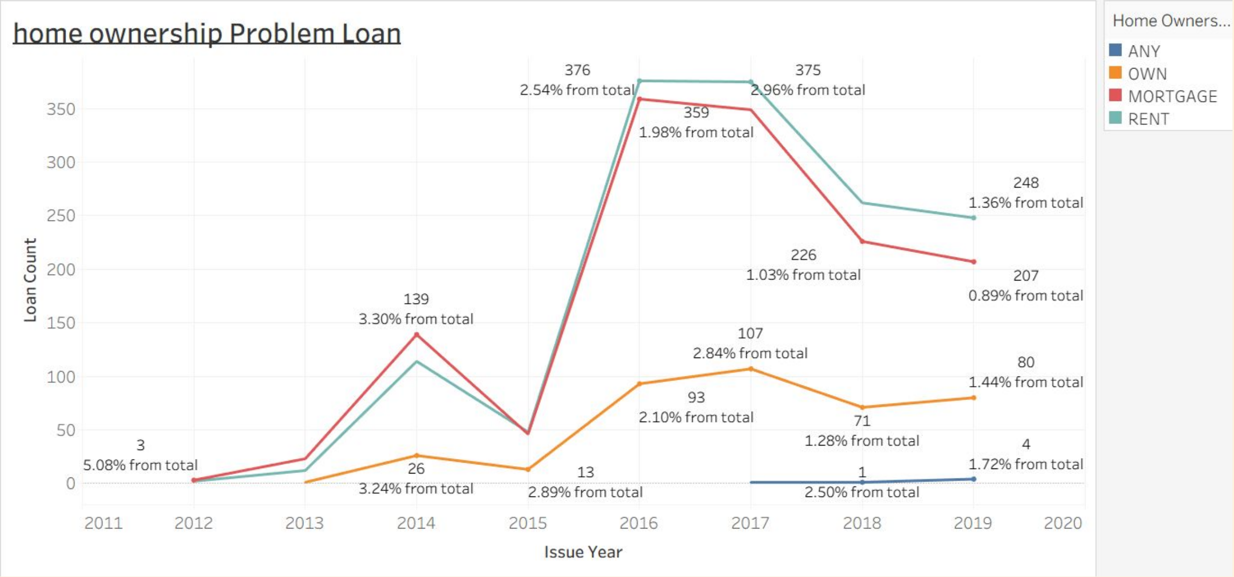
Insights :

- ➢ Mortgage and Rent dominate healthy loan volume across all years, showing these are core performing customer bases.
- ➢ Own and Any segments contribute very small portions of total normal loans, indicating limited portfolio impact.

Recommendation :

- ➢ Prioritize acquisition and retention strategies for Mortgage and Rent customers since they drive portfolio growth.
- ➢ Maintain standard credit policy for Own segment but not primary growth focus.

Home Ownership - Problem Active Loan



Mortgage and Rent cohorts also dominate problem loans across years, indicating risk is exposure-driven rather than segment-driven.

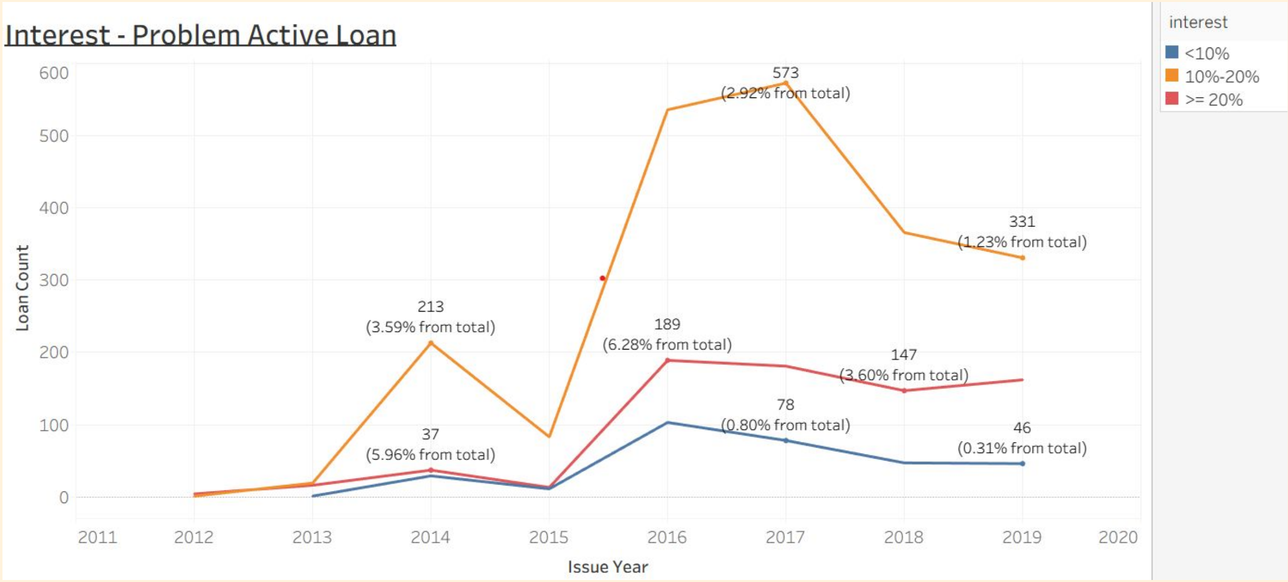
Insights :

- ➢ Problem loans are also highest in Mortgage and Rent — suggesting risk is driven by exposure size, not segment weakness.
- ➢ No strong spike in Own or Any, meaning they are not major risk drivers.

Recommendation :

- ➢ Apply tighter risk monitoring on Mortgage and Rent (early warning, behavior monitoring, income revalidation).
- ➢ Focus collection strategy proportionally to exposure volume, not just risk rate.

Interest Metrics - Normal Active Loan



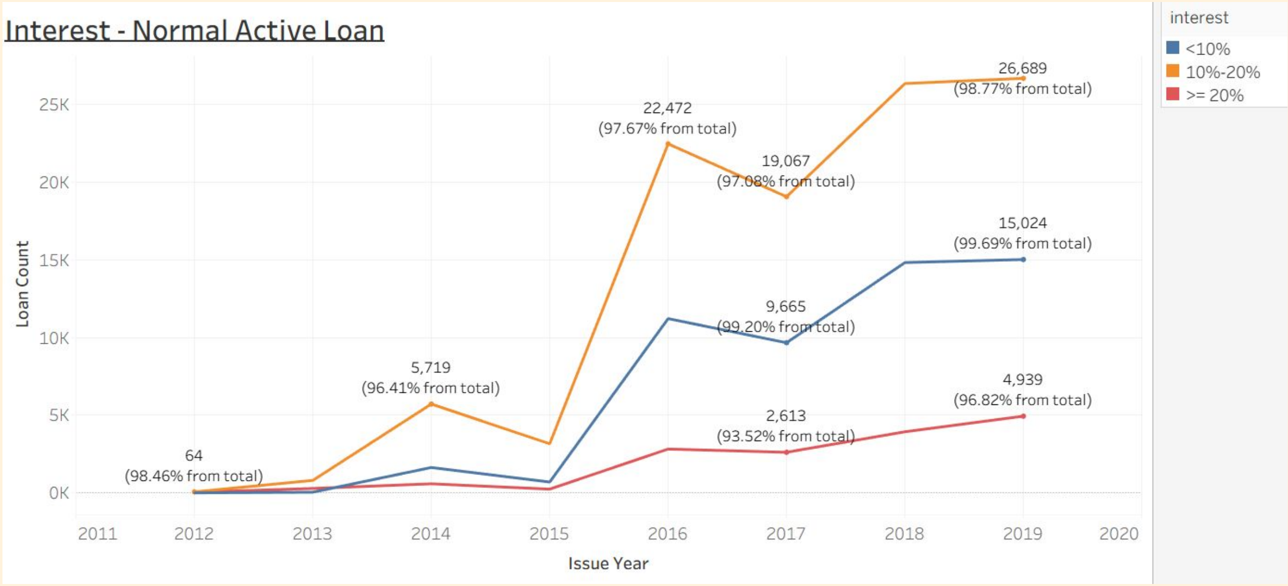
Insights :

- ➢ Normal loans are concentrated in the 10–20% interest cohort, which shows stable growth across issue years and consistently high repayment performance.

Recommendation :

- ➢ Apply tighter risk monitoring on Mortgage and Rent (early warning, behavior monitoring, income revalidation).
- ➢ Focus collection strategy proportionally to exposure volume, not just risk rate.

Interest Metrics - Problem Active Loan



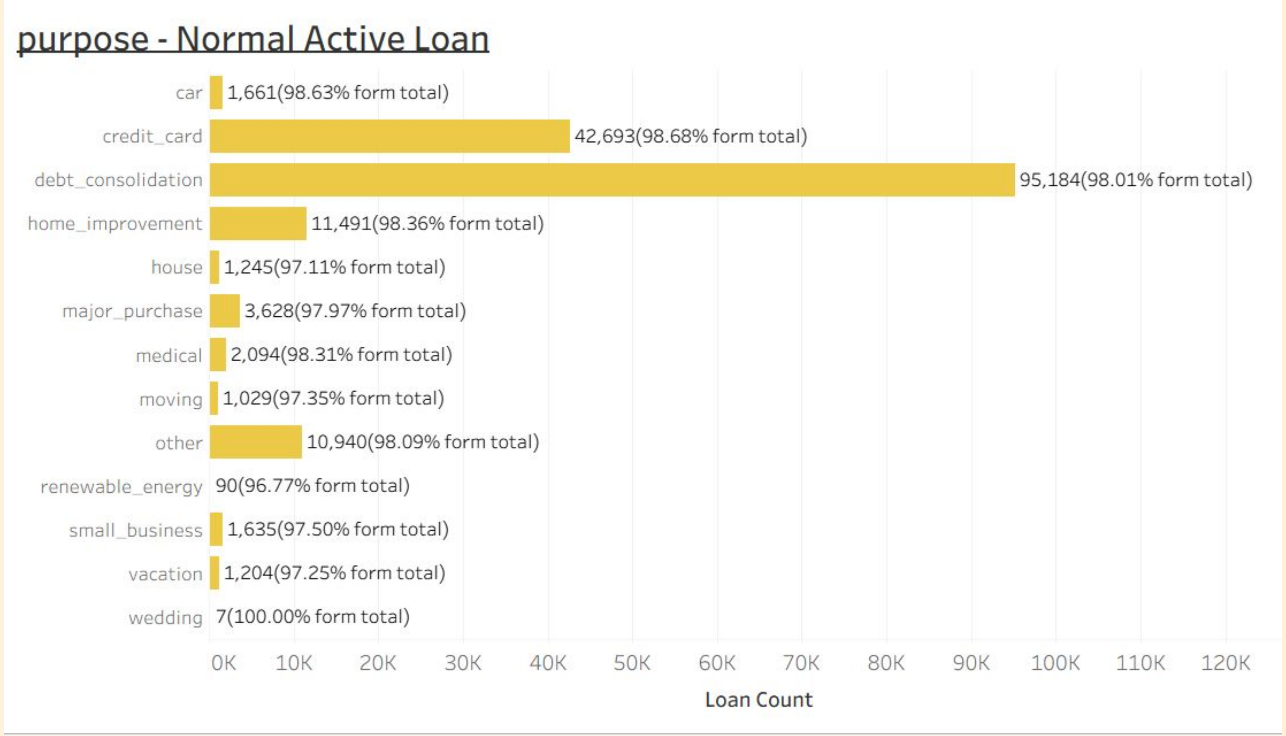
Insights :

➢ Problem loans are disproportionately higher in the ≥20% interest cohort, indicating elevated repayment risk as interest increases.

Recommendation :

➢ Apply stricter approval and monitoring for loans with interest ≥20%, including tighter affordability checks and early collection triggers.

Loan Purpose Metrics - Normal Active Loan



Insights :

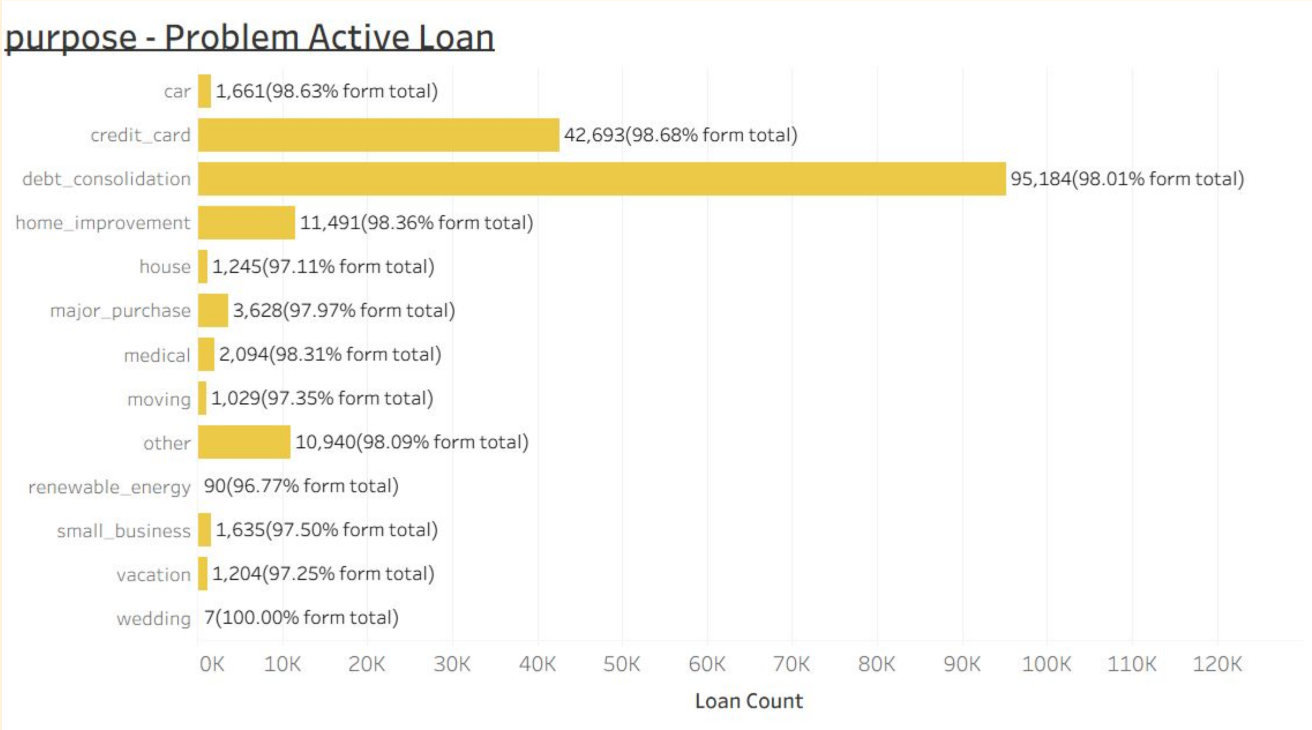
➢ Debt consolidation and credit card purposes dominate normal loan volume, suggesting these purposes align well with borrower repayment capacity.

Recommendation :

➢ Continue prioritizing these purposes as core products, while maintaining current underwriting standards.

- Debt consolidation loans are core to business scale but also risk concentration.
- Niche purposes (renewable energy, moving, vacation) have small volumes but higher default percentages.
- Car and home improvement loans show relatively controlled risk.

Loan Purpose Metrics - Problem Active Loan



Insights :

- ➢ Problem loans are heavily concentrated in high-volume purposes such as debt consolidation and credit cards, driven by exposure rather than niche misuse.

Recommendation :

- ➢ Implement purpose-based risk controls for high-volume categories, such as enhanced income verification and usage monitoring.

- Debt consolidation loans are core to business scale but also risk concentration.
- Niche purposes (renewable energy, moving, vacation) have small volumes but higher default percentages.
- Car and home improvement loans show relatively controlled risk.

Insights

Key Portfolio Insights

1. Credit Quality Improved as the Portfolio Scaled

- Post-2016 cohorts show strong growth with consistently low TKB30, confirming underwriting improvements.
- Recent vintages achieve scale without sacrificing credit quality.

2. Risk Is Concentrated in Specific Segments, Not Systemic

- Problem loans cluster in certain income bands, interest tiers, and purposes, rather than across the whole portfolio.
- This indicates targeted risk issues, not overall portfolio weakness.

3. Customer Profile Strongly Shapes Loan Outcomes

- Mid-to-high income borrowers (≥50K) and homeowners dominate normal loans.
- Lower income borrowers and renters appear more frequently in problem loan cohorts.

4. Pricing and Purpose Amplify Risk When Combined with Volume

- 10–20% interest loans and debt consolidation drive both portfolio growth and risk due to sheer exposure size.
- High interest (≥20%) increases risk, but volume—not rate alone—determines total problem loans.

Insight Summary:

- **RevoFin's portfolio is fundamentally healthy, but risk concentrates where pricing, borrower capacity, and exposure volume intersect.**

Recommendations

Strategic Recommendations

1. Institutionalize Post-2016 Credit Strategy

- Use 2018–2019 cohorts as the benchmark for underwriting, scoring, and approval rules.
- Avoid relaxing standards even as loan volume grows.

2. Apply Segment-Based Underwriting Controls

- Tighten affordability and income validation for:
    - Low-income borrowers Renters
    - Mid-interest (10–20%) loans with large exposure

3. Rebalance Portfolio Growth Toward Low-Risk Segments

- Actively grow:
    - High-income borrowers
    - Homeowners without mortgages
    - Low-interest (<10%) loans
    - Set exposure caps on high-risk, high-volume segments.

Recommendations

4. Refine Risk-Based Pricing & Tenor Strategy

- Reprice or shorten tenors for ≥20% interest and stressed borrower segments.
- Reward strong borrowers with competitive pricing to improve risk-adjusted returns.

5. Strengthen Early Warning & Cohort Monitoring

- Track cohort-level KPIs such as:
- TKB30 and roll rates
- Segment-level delinquency migration
- Intervene early before loans move into late-stage delinquency.
- Recommendation Summary:
- Focus on precision risk management, not broad tightening — protect growth while controlling concentrated risk.

You can check the tableau public here

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appendix-SQL- active loan

```
SELECT l.loan_id,
	l.customer_id,
	c.emp_title,
	c.emp_length,
	c.home_ownership,
	c.annual_inc, c.annual_inc_joint,
	c.verification_status,
	c.zip_code,
	c.addr_state,
	c.avg_cur_bal,
	c.Tot_cur_bal,
	l.loan_status,
	l.loan_amount,
	l.state, sr.subregion,sr.region,
l.funded_amount, l.term,
CAST(REPLACE(l.term, ' months', '') AS INT) AS
term_month,
	l.int_rate,	l.installment, l.grade,
	l.issue_d,	l.issue_date,
	STR_TO_DATE(CONCAT('01 ',issue_date), '%d %M
%Y') issue_dt,
	MONTH(STR_TO_DATE(CONCAT('01 ',issue_date),
'%d %M %Y'))
	issue_month,	l.issue_year,
	l.pymnt_plan, l.type, l.purpose,
	l.description, l.notes
FROM loan l
INNER JOIN customer c ON l.customer_id = c.customer_id
INNER JOIN state_regions sr ON l.state = sr.state
WHERE l.loan_status NOT IN ('Fully Paid','Charged Off')
```

appendix-SQL-cohort yearly

```
WITH RECURSIVE calendar AS (
   SELECT DATE('2012-01-01') AS month_dt
   UNION ALL
   SELECT DATE_ADD(month_dt, INTERVAL 1
MONTH)
   FROM calendar
   WHERE month_dt < CURDATE()),
loan_base AS (
   SELECT   loan_id,
      funded_amount,
      issue_dt,  issue_dt AS amort_dt,
      int_rate/100/12 AS monthly_rate,
      term_month,    installment,
      funded_amount AS remaining_balance,
      0 AS payment_number
   FROM active_loan),
amort AS (  SELECT * FROM loan_base
   UNION ALL
   SELECT  a.loan_id,    a.funded_amount,
 a.issue_dt,   DATE_ADD(a.amort_dt, INTERVAL 1
MONTH),
a.monthly_rate,  a.term_month,   a.installment,
GREATEST( a.remaining_balance -
        (a.installment - a.remaining_balance *
a.monthly_rate),  0 ) AS remaining_balance,
      a.payment_number + 1
  FROM amort a
   WHERE a.payment_number < a.term_month),
loan_month AS (
   SELECT  a.loan_id,
      a.funded_amount,
      a.remaining_balance AS os_amount,
      a.amort_dt,
      YEAR(a.issue_dt) AS cohort_year,
      TIMESTAMPDIFF(MONTH, a.issue_dt, a.amort_dt) AS
month_number,
      l.loan_status
   FROM amort a
   JOIN active_loan l
    ON a.loan_id = l.loan_id),
base AS (
SELECT
      YEAR(issue_dt) AS cohort_year,
      SUM(funded_amount) AS base_fa,
      COUNT(DISTINCT loan_id) AS base_loans
   FROM active_loan
   GROUP BY YEAR(issue_dt))
SELECT
   lm.cohort_year,
   (lm.cohort_year + lm.year_number) AS activity_year,
   Lm.year_number,
COUNT(DISTINCT CASE WHEN lm.is_anniv = 1 THEN
lm.loan_id END) AS cust_active,
SUM(CASE WHEN lm.is_anniv = 1 THEN lm.os_amount END)
AS OS_anniv,
   SUM(CASE WHEN lm.is_anniv = 1 THEN lm.os_amount
END) / b.base_fa AS retention_os,
   SUM(
      CASE   WHEN lm.is_anniv = 1 AND lm.loan_status <>
'Default'   THEN lm.os_amount   END
   ) AS ENR_anniv,
   SUM(
      CASE  WHEN lm.is_anniv = 1 AND lm.loan_status <>
'Default'   THEN lm.os_amount    END
   ) / b.base_fa AS retention_enr,
   SUM(
      CASE
         WHEN lm.is_anniv = 1
          AND lm.loan_status IN ('Late (31-120 days)','Default')
         THEN lm.os_amount
      END
   ) AS TKB30_anniv,
   SUM(
      CASE   WHEN lm.is_anniv = 1
          AND lm.loan_status IN ('Late (31-120 days)','Default')
         THEN lm.os_amount   END
   ) / b.base_fa AS retention_tkb30
FROM (    SELECT  *,
      FLOOR(month_number/12) AS year_number,
      CASE WHEN month_number % 12 = 0 THEN 1 ELSE 0
END AS is_anniv
   FROM loan_month) lm
JOIN base b
   ON lm.cohort_year = b.cohort_year
GROUP BY lm.cohort_year, (lm.cohort_year +
lm.year_number), lm.year_number
ORDER BY lm.cohort_year, lm.year_number;
```